Information-theoretic Frontier Selection for Environment Exploration

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***Abstract*— The exploration of unknown environments using autonomous mobile robots is essential for different applications, for example, search and rescue missions. The main objective is to efficiently transverse the environment and build a complete and accurate map. However, different applications may demand different exploration strategies. The simplest strategy is a greedy approach which visits the closest frontier without considering if it will yield a significant reduction in map uncertainty. In this paper, we propose a novel method to predict information beyond the candidate frontiers by analyzing the local structure. Next, the utility function chooses a candidate locations using Shannon entropy. The methodology was evaluated through several experiments in a simulated environment, showing that our exploration approach is better suited for rapid exploration than the classic Near-Frontier Exploration (NFE).**

I. Introduction

The task of exploring unknown environments using autonomous mobile robots is essential for many applications. Basically, the goal is to perceive the environment with the robot's sensors and build a complete and accurate map. However, different applications demand different exploration strategies. For example, rescue robots are required to act quickly, covering the environment as much as possible in a short amount of time, while other applications demand map quality without stringent time constraints.

In general, an autonomous exploration strategy can be divided in three basic steps: (i) planning the next best viewpoint, (ii) navigating to the target selected, and (iii) updating the map and robot models. In most cases, a strategy can be characterized mainly by the first step, which is responsible for the robot's decision about where to go next in order to enhance the map's information. This phase analyzes and evaluates a subset of locations in order to select the next best destination for the robot to explore.

One of the most well-known exploration methods was presented by Yamauchi [1] and focuses on visiting map frontiers. Frontiers are defined as the boundaries between the known and unknown areas in a map. In [1], the strategy presented always selects the closest frontier for exploration. Despite its simplicity, the Near-Frontier Exploration (NFE) is often not the best approach in scenarios where time is critical as it drives the robot to the location of the closest frontier without considering if it will yield a significant reduction in map uncertainty. Considering this, information-based approaches have been used in recent years to better evaluate the next best destinations for short time explorations [2], [3], [4].

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Therefore, given the robot's position and the current map, the main problem is to determine the next exploration destinations in order to efficiently map the environment (Figure 1). In this manner, we propose a utility function to evaluate candidate locations using Shannon entropy [5]. For each candidate frontier, it calculates the current entropy of the map and the estimated entropy based on the belief about the map configuration after a robot visits this destination. The difference between these entropies is called Mutual Information or Information Gain. However, it is not a simple task to predict the unseen area and it is not computationally feasible to calculate all potential observations before the robot reaches a candidate location.

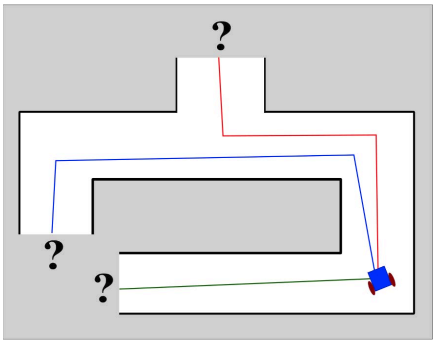

Fig. 1: In exploration tasks, a fundamental problem is the determination of next exploration destinations in order to maximize the environment coverage.

In this paper we consider a simple real-world assumption to make predictions possible: most indoor environments are composed of straight walls with intersections. Exploiting these characteristics, we propose a novel method to anticipate information beyond the candidate frontiers by analyzing local structure around each frontier and predicting how it should propagate. Using this technique, it becomes viable to calculate the information gain and to select the next best destination for the robot. In this way, the region with the highest map uncertainty will attract the robot and the environment will be covered faster making our exploration approach better suited for rapid mapping of an unknown environment than classic Near-Frontier Exploration.

II. Related Work

Autonomous exploration is of great importance, and has been the goal of different research fields, especially Robotics. It is a non-trivial task that can be solved by many distinct approaches. In this paper, we focus on strategies that can map the largest area in the shortest amount of time.

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Among the first autonomous exploration strategies were those based on fixed and random trajectories [6], [7]. While offering low complexity, these classical methods were rather rudimentary and thus efforts were made to improve the decision making of the robots. One such approach takes into account the cost of reaching candidate destinations [1]. Others attempt to further reduce exploration time by devising utility functions that estimate not only the cost of visiting a particular goal but also the expected information gain therein [2], [3], [4].

The Information Gain has been used in many different ways and can be grouped into three main categories: (i) map approaches that only use the information obtained from the map to calculate the levels of uncertainty [8], [9]; (ii) geometric approaches that use the robot's sensor model and map to predict the areas beyond the frontiers [10], [11]; and (iii) information-theoretic approaches that use the robot's sensor model, map and sometimes environment knowledge to predict information beyond the frontiers [12], [13].

Information-theoretic strategies have proved to be a better alternative for calculating information gain. A few such methods guide the robots to frontiers that potentially maximize the reduction of the map's uncertainty. One technique for doing this is to use knowledge of previously seen environments to predict and, consequently, to better estimate the information gain. For example, [13] uses a method that tries to match the area beyond the frontiers with local known maps. A database of local maps is incrementally built during exploration with the best match indicating the best prediction. This method yields good results for areas with repetitive structure but can suffer in more unstructured environments.

Other approaches incorporate environmental knowledge in the decision making process of the robots in order to reduce exploration time. For example, [14] uses knowledge about the structure of the environment to speed up the exploration by identifying corridors and giving them higher utility values, thus encouraging the robot to move through corridors first and then into adjoining rooms. [15] uses knowledge about the environment to coordinate a group of robots during exploration. However, these techniques do not use information gain for its utility function.

For a more in depth discussion related to different autonomous exploration strategies we refer the reader to [16], which presents a recent overview of the main challenges and existing techniques to the problem.

Our main contribution in this paper is the development of a simple yet novel technique to calculate the uncertainty beyond the frontiers. We focus on the uncertainty terms of the utility functions leaving out other auxiliary costs such as travel distance.

III. Methodology

In this section, we detail the two main components of our proposal: the information-theoretic frontier selection approach and the method for predicting areas beyond frontiers. The first subsection briefly describes the technique employed to determine the information gain of potential robot actions, while the second part presents the approach developed to predict information yet to be collected. When combined, these approaches offer a better strategy to select the best frontiers during exploration under time constraints.

A. Background

The exploration problem can be also classified as a mapping task on which the robot needs to generate a representation of the perceived environment. In this work, we used the Occupancy Grid model to build the maps.

An Occupancy Grid Map consist in a spatial 2D grid that reflects the occupancy of the environment. Based on the observations (z) and actions (x) made by the robot from initial pose till the last pose, the map (M) is given by

$$p(M|z_{1:t}, x_{1:t}), \tag{1}$$

where each cell of the grid is composed by a binary random variable which indicates if the cell is occupied "1" or free "0".

The concept of a frontier was introduced by Yamauchi [1]. In a occupancy grid map, a frontier is composed by all cells that are on the boundary between the free and unknown areas. In this work, we define the size of a frontier as the distance between the two extremities cells in additional to some extra distance to cover the walls that limit the frontier.

B. Information-Theoretic Frontier Selection

The autonomous exploration problem can be described as follows: at each timestep, the robot must decide which action to execute, for example, where to go next. During navigation to the selected location, new observations are made and new information is integrated to the map. Once the robot reaches the destination, a new decision needs to be made about where to go next. In order to reduce the time it takes to complete the exploration process, the estimated information gain is employed as the method by which goals points are chosen.

In other words, assuming the robot's pose is known, we are interested in determining which action will most quickly reduce the map's uncertainty. If the estimate information gain of one frontier is higher than the others then there is a belief that the free area beyond the frontier will yield the higher aggregation of new information. Thus, navigating to this frontier should be the best action for the robot at that moment.

The estimated information gain, also called mutual information, can be defined as the expected change in entropy in the belief about the robot's observations Z and the map M, given by

$$I(M; Z_{t+1}|z_{1:t}) = H(M|z_{1:t}) - H(M|Z_{t+1}, z_{1:t}), \tag{2}$$

where $z_{1:t}$ represents the observations already made and Z_{t+1} is the predicted observation given a specific robot action to be executed. In this equation, there are two entropy values to be calculated: the current entropy of the map conditioned on all measurements and the future or estimate entropy of the map conditioned on all measurements and an expected future sensor reading. To calculate the current entropy of the map conditioned on all sensor readings the following equation is used [17]:

$$H(M|z_{1:t}) = -\sum_{c \epsilon M} p_c \log p_c + (1 - p_c) \log(1 - p_c), \quad (3)$$

where c is the Bernoulli random variable associated with a cell of the occupancy grid and p_c is the probability of the cell to be occupied. The values of p_c range from 0, representing absolute certainty of that cell being free (white on an Occupancy Grid Map), to 1, meaning absolute certainty of being occupied (black on an Occupancy Grid Map). A value of 0.5 at the middle of the range (gray on an Occupancy Grid Map) denotes a cell whose state is uncertain.

Unfortunately, it is not feasible to calculate the posterior entropy without making a simplification or a prediction [13]. This is expected since there exist a vast number of potential measurements for the observation Z_{t+1} which grows exponentially with the dimension of the environment over time.

In this work, we propose a suitable approximation that depends only on the current map. With the right predictions, it becomes possible to calculate the posterior entropy and then the utility function (Equation 4). As was mentioned before, the utility function will select the best action **a**, according to (Equation 5), that will potentially lead the robot to maximize the uncertainty reduction of the map based on the estimated information gain.

$$U(a) = I(a), \quad (4)$$

$$a^* = \underset{a \epsilon A}{\text{argmax}}\{I(a)\}. \quad (5)$$

C. Undiscovered Areas Prediction

In order to make the prediction, it is assumed based on common observation that indoor environments can be mostly characterized by straight walls with intersections, e.g, corridors and rooms. In this scenario, there are essentially two ways walls can be expected to propagate beyond the frontier: in a straight line or a corner. By applying this idea to all frontiers, it is possible to predict the new free and non-free areas of the map.

However, the space downstream of frontiers contain some combination of straight segments and corners and thus the further a prediction extends beyond the frontier the less likely it is to resemble the true state of the environment. Therefore, this prediction can not be extended to big areas since it would probably create a false certainty after some area covered.

In order to remedy this issue we create limitation zones for these predictions by restricting the area covered by the prediction based on the frontiers' size. In other words, the area covered by the predicted propagation of wall segments is proportional to the size of the frontier.

Finally, to determine the direction in which the wall is most likely to propagate, the local area around the frontier is analyzed. Each cell that is non free will be classified as part of a wall. After detecting the walls, the pixels that represent the free space are analyzed. The walls bordering unknown territory are extended. If the free cells block the propagation of the wall it means the wall needs to turn perpendicular to the direction of the unknown area. Afterwards, the free cells are plotted using a wave propagation method. The cells in contact with the frontier but with no information are updated to free. Then, the new free cells become the frontier and the process is repeated till the new frontier cells reach the limits of the prediction zone. The predictions are made with a linear reduction of the certainty to avoid big propagation in big zones. For a better understanding, Figure 2 shows a few examples of predictions during an exploration experiment.

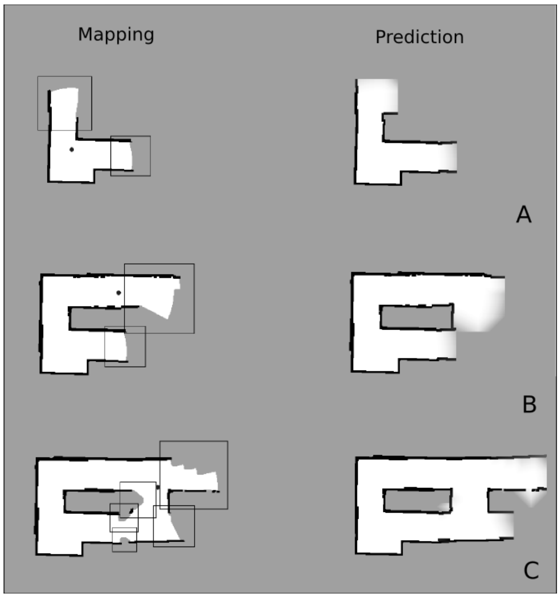


Fig. 2: Exploration steps. The left column represents the mapping steps and the right column show the prediction based on the structure of the walls around each frontier. The squares covering the frontier indicate the extents of the prediction zones.

As can be seen from Figure 2, the prediction zones are bigger when the size of the frontiers are bigger and vice versa. In Figure 2.A, one wall propagates straight, hits a free cell, and turns right while the others stay straight till they encounter the limits of the prediction zone. In the second step (Figure 2.B), the difference is size of the frontiers and subsequently the prediction zones can clearly be seen. In the same step, the reduction in the certainty of the prediction can also be clearly seen. Finally, the last step (Figure 2.C) demonstrates the fact that even when there exists a cluster of frontiers in a small area, the method can still predict the behavior of the walls and select the frontier with the highest estimate information gain.

D. Environment Exploration

The pseudocode of the proposed methodology for better understanding of the whole exploration process is presented in Algorithm 1.

Algorithm 1 InformationGainExploration()

```
for each timestep t do
    d ← checkDestination()
    m ← updateMap()
    p ← updateRobotLocation()
    if isDestinationReached(p, d) then
        e ← calcEntropy(m)                          ▷ Eq. (3)
        f[] ← checkFrontiers(m)
        if f.size() == 0 then
            finishExploration()
        end if
        occCells[] ← selectOccupiedCells(m)
        s[] ← createLineSegments(occCells[])
        for i = 1 to f.size() do
            fSize ← calculateFrontierSize(f[i])
            w ← createWindow(f[i], fSize, m)
            //Detect the walls connected to frontier
            wallsF[] ← checkWalls(f[i], s[], m)
            pMap ← expandWalls(wallsF[])
            pMap ← expandFreeCells(w, f[i], pMap)
            pe ← calcEntropy(pMap)                  ▷ Eq. (3)
            ig[i].add(calculateIG(e, pe))           ▷ Eq. (2)
        end for
        d ← checkForNextDestination(ig[])           ▷ Eq. (5)
        path ← newPath(p, d, m)
    end if
end for
```

The algorithm above describes the steps necessary to execute an exploration task. For each timestep, the robot checks if the current goal waypoint has been reached. If it has, the robot calculates the next best destination; otherwise the robot keeps navigating till stoping in the desired location. To calculate the new target, it is necessary to select each frontier from the map and calculate the information gain using the prediction technique. Afterwards, the frontier with the highest information gain is selected as the new destination.

After selecting the next destination to be explored, a new path is generated. In this work, we used a basic technique to generate the paths, it is known as the Wavefront Propagation algorithm. Basically, the wavefronts propagate from the robot's position cell till reaches the destination cell. Each new wavefront generated is marked with a higher value than the previous. The shortest path can be created by tracing the highest descent of wavefronts from the destination back to the source.

IV. Experiments

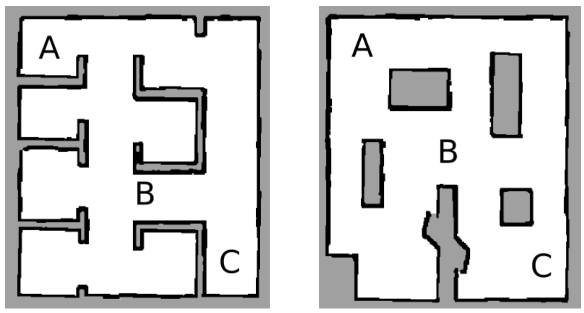
In this section we present an experimental evaluation of the methodology in order to assess the advantages of our predictive exploration approach and support our claim that estimating the information gain for each candidate destination it is possible to cover the unknown environment faster than classic frontier-based approaches.

We performed a series of experiments in two simulated scenarios, a cave-like and an office environment. These environments were chosen because they have characteristics common to most indoor spaces. The code was written in C++ using ROS (Robot Operating System) and executed on a computer running 64-bit Ubuntu 14.04 with an Intel i7 processor and 8GB of RAM. The ROS gmapping SLAM package was used to provide the robot's location and the parameters of the sensors were set with minimum error.

A. Experimental Procedure

To evaluate the proposed algorithm through simulation, three trials are run in each environment with the robot's initial position being randomly chosen for each trial. Once started, the robot proceeded to perceive the environment and identify candidate frontiers. The proposed utility function then selected the next best frontier to maximize environment coverage. This process was repeated until no more frontiers were detected.

Figure 3 presents the ground-truth maps obtained by manually navigating the robot in both scenarios, as well as the initial positions (marked as A, B and C). The left side shows an office with some rooms and corridors and the right side shows a cave with large open areas containing some rocks and columns.



(a) Office (b) Cave

Fig. 3: The final 2D occupancy-grid maps (20x20 m^2) for each environment used in the simulations.

To better evaluate the proposed approach discussed in the methodology, we initially allotted 10 minutes for the robot to complete the exploration task. This initial time was used to compare the proposed algorithm and the traditional NFE approach when subjected to the same time limitation. After the first 10 minutes, the robot continued the exploration process until the whole environment was covered. In the end, the total time required to complete the exploration mission and the distance traveled by the robot were compared to the baseline results.

Some common metrics, such as completeness and quality of a map [18], were not applied in the experiments because of the high precision of the robot's location. In this work, all the experiments concluded with a high accuracy of the map with completeness. For these reasons, it is unnecessary to plot charts comparing such information.

B. Presentation of the Results

We compared our exploration strategy with the classic frontier-based technique known as Near-Frontier Exploration (NFE) [1]. NFE was adopted as a baseline because of its established simplicity and good performance. However, NFE does not contemplate any form of uncertainty reduction. Therefore, it often does not select the areas with the biggest information gain to rapidly maximize the map coverage.

The exploration results related to the initial 10 minutes of exploration are shown in Figure 4. The graphics present the percentage of the map covered versus the exploration time. As it was mentioned before, we set three different initial poses for the robot. Basically, the positions were defined randomly in three different regions of the map: the two extremities and the center.

As can be seen from Figure 4, a clear improvement of the proposed strategy when compared to the NFE approach is obtained for most cases. Our prediction technique makes it possible for the robot to select the areas with the biggest estimated information gain. Thus, the robot moves through the environment revealing large areas while leaving small parts unseen. Most of these small portions of the map correspond to irrelevant fragments of the environment such as room corners. In this way, during the critical first minutes of exploration, the map built using our technique contains the larger more important areas, representing a higher knowledge of the environment in less time. Figure 5 illustrates the path executed by the robot and the map obtained.

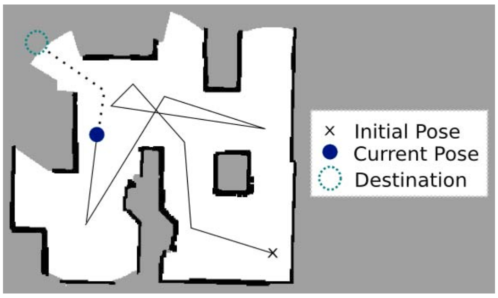
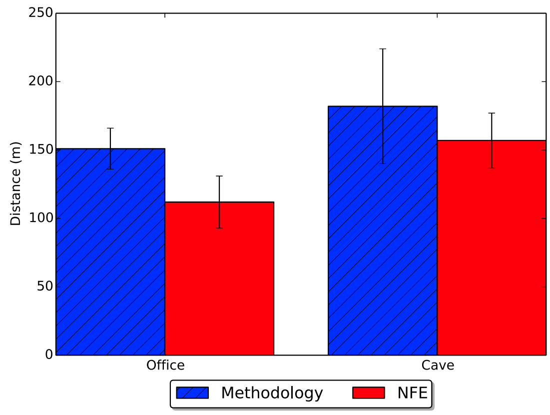


Fig. 5: Example of the path traveled by the robot (solid line) and the map generated after 10 minutes of exploration.

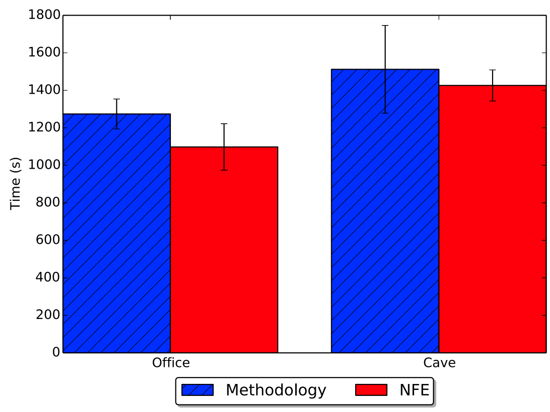
For both environments, the proposed strategy demonstrated a similar response. It confirms that the prediction technique demonstrated in this paper brings good results for indoor environments. But it is important to remember the influence of the prediction in the robot's decision making. The predictions are made in a local part of a map and highly influenced by the size of each frontier. It is possible for the robot to select a frontier with the highest estimated information gain that leads it to an undesired area. In addition, the prediction may not always accurately represent reality. Nevertheless, the results show that while not always accurate, the prediction used to inform our information-bases algorithm positively improved the robot's decision making.

However, after the initial 10 minutes of exploration, the rate of information gain decreased and the percentage of the environment covered becomes similar to that of the NFE approach. As mentioned before, the information gain method leaves some areas to be explored later as opposed to the NFE strategy which explores all the closest areas first before advancing. Because of this, the proposed approach causes the robot to return to previously visited areas in order to complete the map. This characteristic can also be observed in Figure 5.

Finally, Figure 6 shows the average total exploration time and distance traveled to completely explore the environments in all experiments. Because of the gaps cited above, the new approach indicates a small increase in both metrics compared to the baseline.



(a) Distance

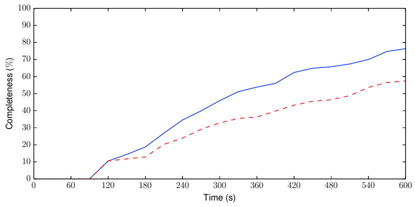


(b) Time

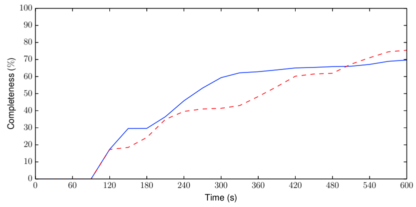
Fig. 6: Comparison between both strategies of the total time spent and distance traveled to complete exploration missions.

V. Conclusion and Future Work

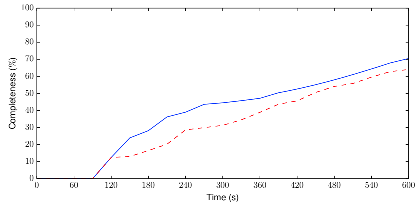
In this paper, we proposed a novel solution for autonomous exploration tasks with time constraints. The strategy combines prediction techniques with an information-theoretic approach in order to enhance rapid environmental coverage. In our approach we exploit some common indoor characteristics to make predictions possible. This technique allows the robot



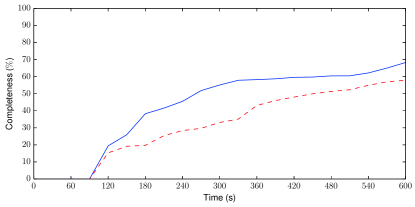
(a) Office (Position A)



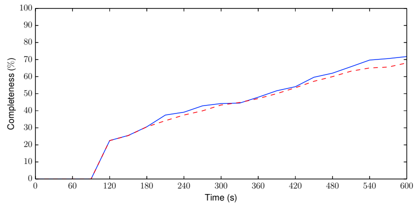
(b) Office (Position B)



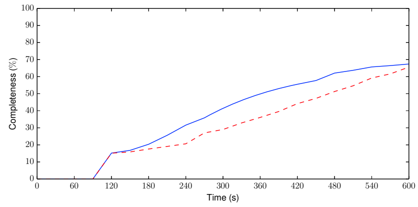
(c) Office (Position C)



(d) Cave (Position A)



(e) Cave (Position B)



(f) Cave (Position C)

Fig. 4: Percentage of the map covered against exploration time (first 10 minutes) for each scenario and initial position. The solid blue line represents the proposed methodology, and the dashed red line the NFE approach.

to calculate the potential information gain for each candidate frontier and make better decisions about where to go next.

In simulation we evaluated the performance of our algorithm using three main metrics: total time, distance traveled and efficiency. The results of the proposed strategy illustrated a better performance compared to the Nearest-Frontier Exploration strategy. As a result, with the proposed approach it becomes possible to obtain a more complete map in a shorter amount of time.

For future work, we intend to develop a new method using a multi-criteria decision maker in order to reduce the total time and distance required to explore the environment completely and improve the first minutes of coverage; areas with the largest information gain will also be judged by other criteria such as cost to reach that region of the map.

Acknowledgments

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